

Calculation of Electric Field and Forces

EXAMPLE 21.4 Vector addition of electric forces in a plane

Two equal positive charges $q_1 = q_2 = 2.0 \mu\text{C}$ are located at $x = 0, y = 0.30 \text{ m}$ and $x = 0, y = -0.30 \text{ m}$, respectively. What are the magnitude and direction of the total electric force that q_1 and q_2 exert on a third charge $Q = 4.0 \mu\text{C}$ at $x = 0.40 \text{ m}, y = 0$?

1. || What are the strength and direction of the electric field at the position indicated by the dot in **FIGURE EX23.1**? Specify the direction as an angle cw from vertical.

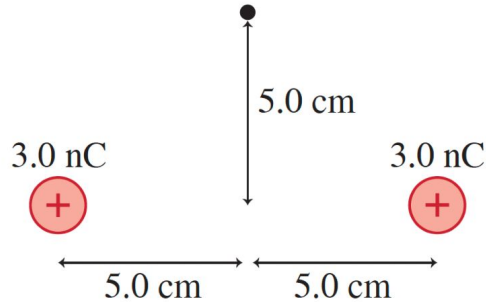


FIGURE EX23.1

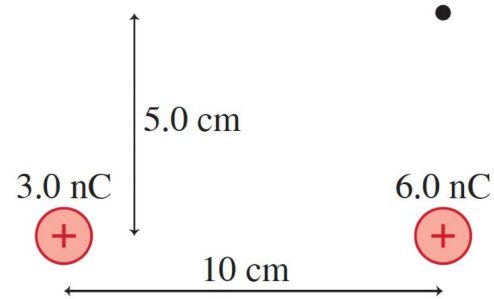
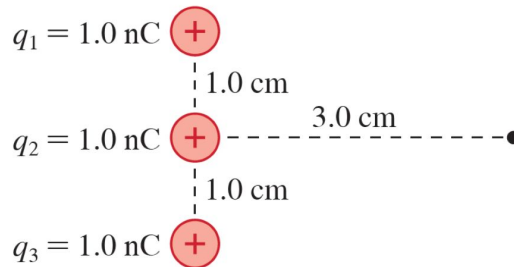


FIGURE EX23.2

2. || What are the strength and direction of the electric field at the position indicated by the dot in **FIGURE EX23.2**? Specify the direction as an angle cw from vertical.

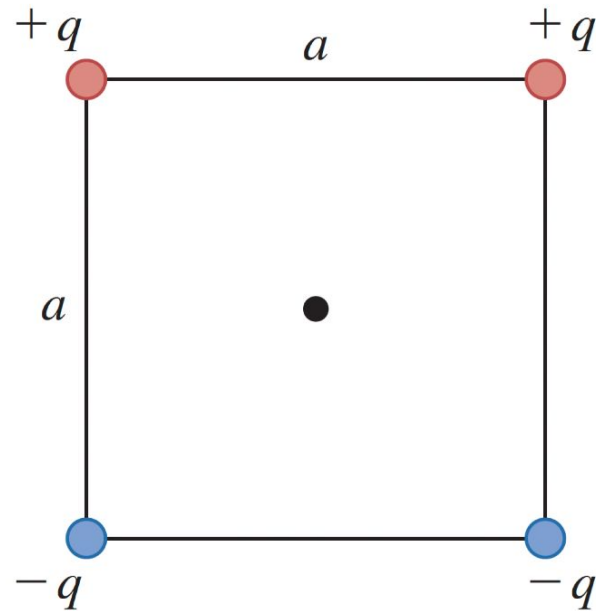
66. | Three 1.0 nC charges are placed as shown in **FIGURE P22.66**. Each of these charges creates an electric field $\vec{E}$ at a point 3.0 cm in front of the middle charge.

- What are the three fields $\vec{E}_1$, $\vec{E}_2$, and $\vec{E}_3$ created by the three charges? Write your answer for each as a vector in component form.
- Do you think that electric fields obey a principle of superposition? That is, is there a “net field” at this point given by $\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$? Use what you learned in this chapter and previously in our study of forces to argue why this is or is not true.
- If it is true, what is $\vec{E}_{\text{net}}$?

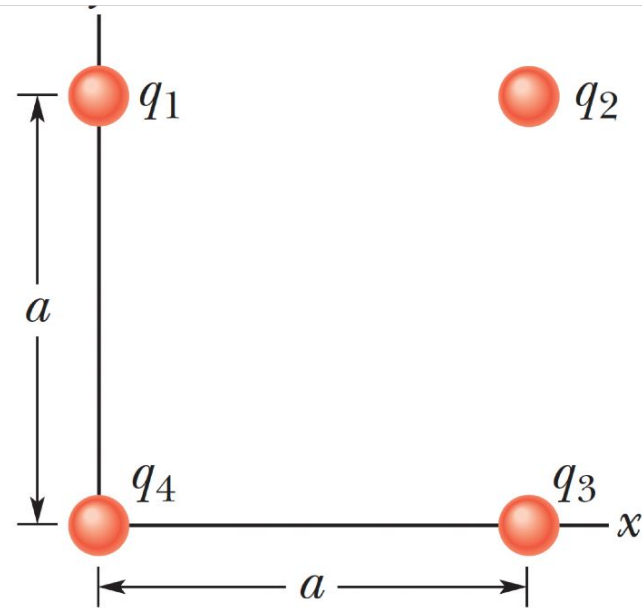


21.38 ●● A point charge is placed at each corner of a square with side length a . All charges have magnitude q . Two of the charges are positive and two are negative (**Fig. E21.38**). What is the direction of the net electric field at the center of the square due to the four charges, and what is its magnitude in terms of q and a ?

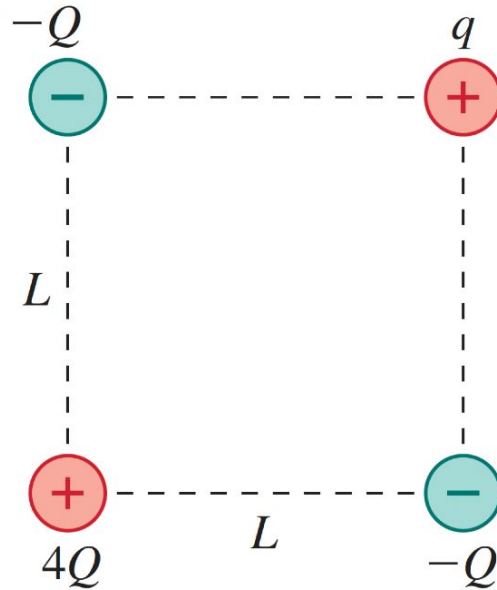
Figure **E21.38**



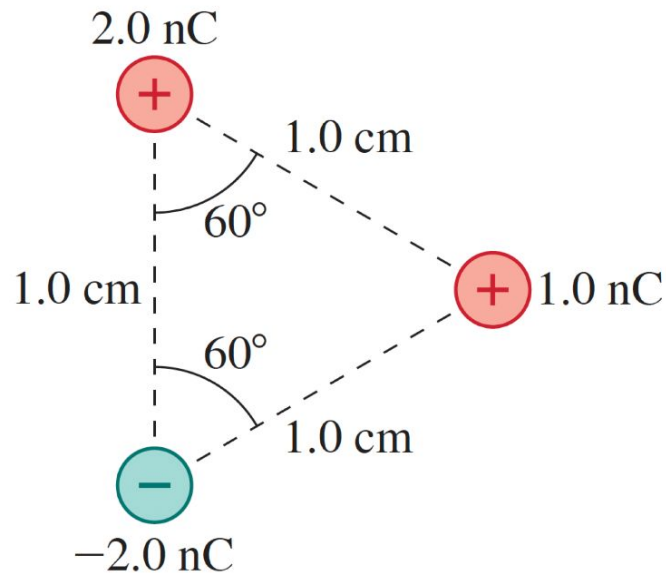
••7 **SSM** **ILW** **WWW** In Fig. 22-35, the four particles form a square of edge length $a = 5.00$ cm and have charges $q_1 = +10.0$ nC, $q_2 = -20.0$ nC, $q_3 = +20.0$ nC, and $q_4 = -10.0$ nC. In unit-vector notation, what net electric field do the particles produce at the square's center?



53. || **FIGURE P22.53** shows four charges at the corners of a square of side L . What is the magnitude of the net force on q ?



19. **II** What is the force $\vec{F}$ on the 1.0 nC charge in **FIGURE EX22.19**? Give your answer as a magnitude and a direction.



•**24** Two tiny, spherical water drops, with identical charges of $-1.00 \times 10^{-16} \text{ C}$, have a center-to-center separation of 1.00 cm. (a) What is the magnitude of the electrostatic force acting between them? (b) How many excess electrons are on each drop, giving it its charge imbalance?

•**25** **ILW** How many electrons would have to be removed from a coin to leave it with a charge of $+1.0 \times 10^{-7} \text{ C}$?

•**26** What is the magnitude of the electrostatic force between a singly charged sodium ion (Na^+ , of charge $+e$) and an adjacent singly charged chlorine ion (Cl^- , of charge $-e$) in a salt crystal if their separation is $2.82 \times 10^{-10} \text{ m}$?

Cancellation of Electric Field and Forces

50. || Charge q_2 in **FIGURE P22.50** is in equilibrium. What is q_1 ?

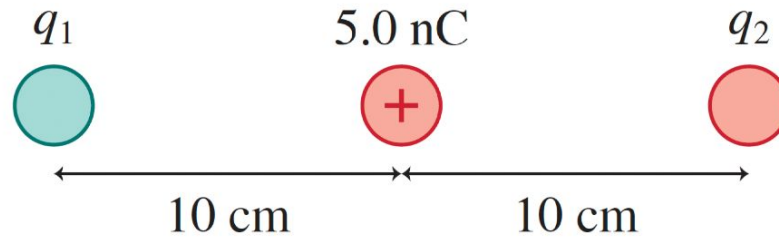
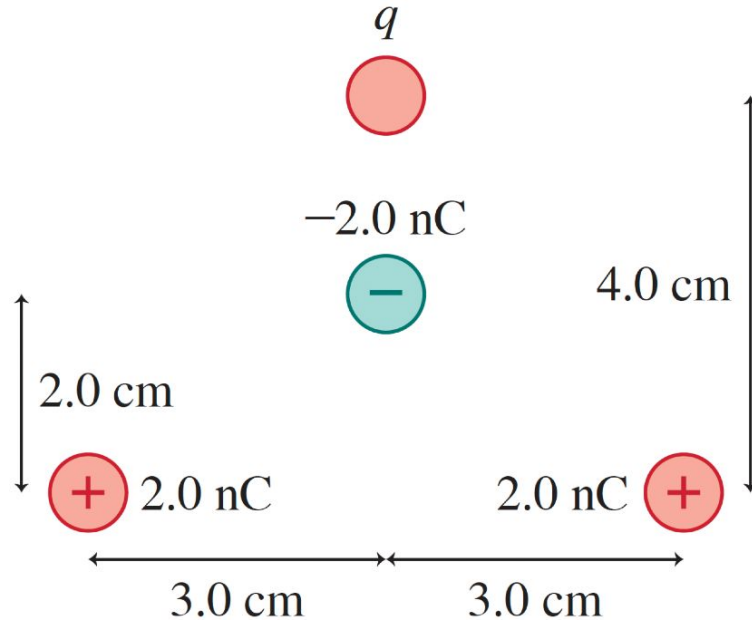


FIGURE P22.50

51. || The -2.0 nC charge in **FIGURE P22.51** is in equilibrium. What is q ?



••10 **GO** In Fig. 21-25, four particles form a square. The charges are $q_1 = q_4 = Q$ and $q_2 = q_3 = q$. (a) What is Q/q if the net electrostatic force on particles 1 and 4 is zero? (b) Is there any value of q that makes the net electrostatic force on each of the four particles zero? Explain.

••11 **ILW** In Fig. 21-25, the particles have charges $q_1 = -q_2 = 100 \text{ nC}$ and $q_3 = -q_4 = 200 \text{ nC}$, and distance $a = 5.0 \text{ cm}$. What are the (a) x and (b) y components of the net electrostatic force on particle 3?

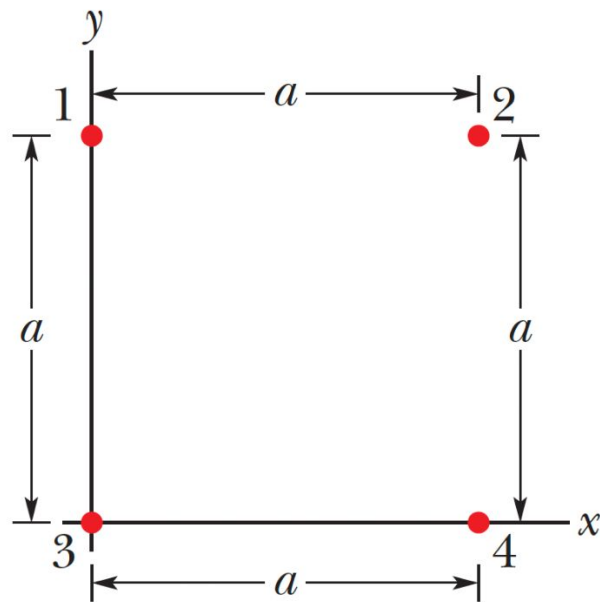


Figure 21-25
Problems 10, 11, and 70.

21.67 ●● Two particles having charges $q_1 = 0.530 \text{ nC}$ and $q_2 = 8.30 \text{ nC}$ are separated by a distance of 1.17 m . At what point along the line connecting the two charges is the total electric field due to the two charges equal to zero?

21.68 ●● A -3.00 nC point charge is on the x -axis at $x = 1.20 \text{ m}$. A second point charge, Q , is on the x -axis at -0.600 m . What must be the sign and magnitude of Q for the resultant electric field at the origin to be (a) 45.0 N/C in the $+x$ -direction, (b) 45.0 N/C in the $-x$ -direction?

21.69 ●●● A charge $+Q$ is located at the origin, and a charge $+4Q$ is at distance d away on the x -axis. Where should a third charge, q , be placed, and what should be its sign and magnitude, so that all three charges will be in equilibrium?

Motion of Charged Particles in Electric Field

••53 GO Two large parallel copper plates are 5.0 cm apart and have a uniform electric field between them as depicted in Fig. 22-60. An electron is released from the negative plate at the same time that a proton is released from the positive plate. Neglect the force of the particles on each other and find their distance from the positive plate when they pass each other. (Does it surprise you that you need not know the electric field to solve this problem?)

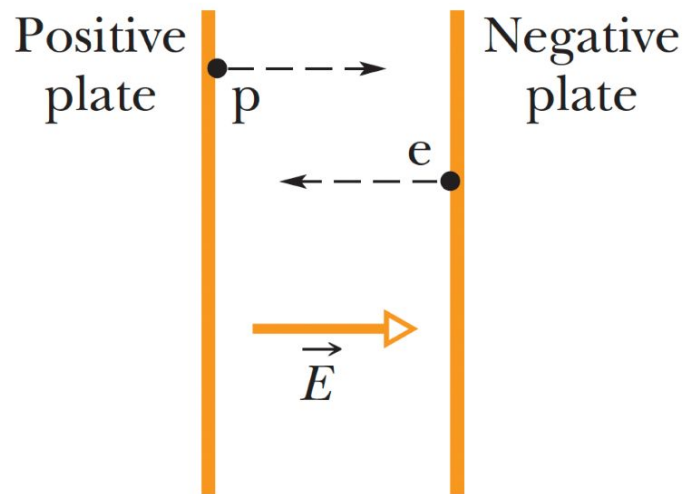


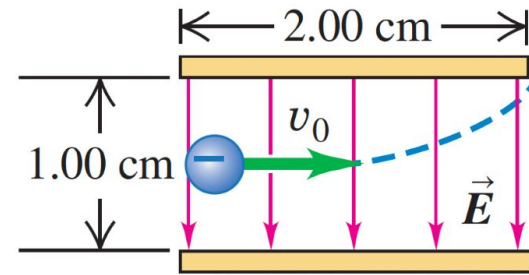
Figure 22-60 Problem 53.

••49 A 10.0 g block with a charge of $+8.00 \times 10^{-5} \text{ C}$ is placed in an electric field $\vec{E} = (3000\hat{i} - 600\hat{j}) \text{ N/C}$. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the electrostatic force on the block? If the block is released from rest at the origin at time $t = 0$, what are its (c) x and (d) y coordinates at $t = 3.00 \text{ s}$?

••50 At some instant the velocity components of an electron moving between two charged parallel plates are $v_x = 1.5 \times 10^5 \text{ m/s}$ and $v_y = 3.0 \times 10^3 \text{ m/s}$. Suppose the electric field between the plates is uniform and given by $\vec{E} = (120 \text{ N/C})\hat{j}$. In unit-vector notation, what are (a) the electron's acceleration in that field and (b) the electron's velocity when its x coordinate has changed by 2.0 cm?

21.27 •• CP An electron is projected with an initial speed $v_0 = 1.60 \times 10^6 \text{ m/s}$ into the uniform field between two parallel plates (**Fig. E21.27**). Assume that the field between the plates is uniform and directed vertically downward and that the field outside the plates is zero. The electron enters the field at a point midway between the plates.

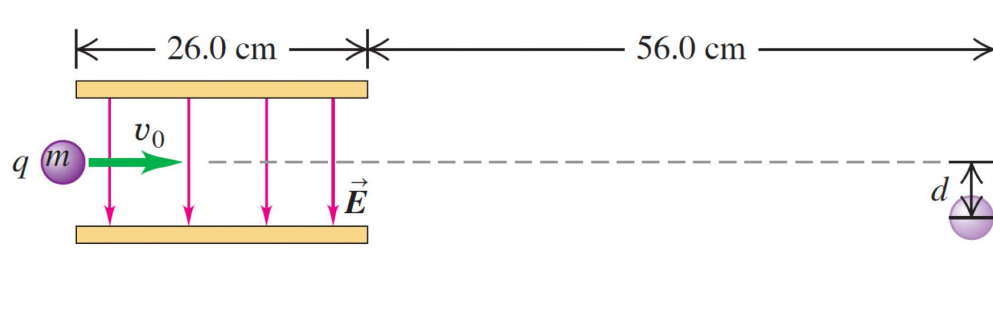
Figure **E21.27**



The electron enters the field at a point midway between the plates. (a) If the electron just misses the upper plate as it emerges from the field, find the magnitude of the electric field. (b) Suppose that the electron in Fig. E21.27 is replaced by a proton with the same initial speed v_0 . Would the proton hit one of the plates? If not, what would be the magnitude and direction of its vertical displacement as it exits the region between the plates? (c) Compare the paths traveled by the electron and the proton, and explain the differences. (d) Discuss whether it is reasonable to ignore the effects of gravity for each particle.

21.78 ●●● A small object with mass m , charge q , and initial speed $v_0 = 5.00 \times 10^3$ m/s is projected into a uniform electric field between two parallel metal plates of length 26.0 cm (**Fig. P21.78**). The electric field between the plates is directed downward and has magnitude $E = 800$ N/C. Assume that the field is zero outside the region between the plates. The separation between the plates is large enough for the object to pass between the plates without hitting the lower plate. After passing through the field region, the object is deflected downward a vertical distance $d = 1.25$ cm from its original direction of motion and reaches a collecting plate that is 56.0 cm from the edge of the parallel plates. Ignore gravity and air resistance. Calculate the object's charge-to-mass ratio, q/m .

Figure **P21.78**



••54  In Fig. 22-61, an electron is shot at an initial speed of $v_0 = 2.00 \times 10^6$ m/s, at angle $\theta_0 = 40.0^\circ$ from an x axis. It moves through a uniform electric field $\vec{E} = (5.00 \text{ N/C})\hat{j}$. A screen for detecting electrons is positioned parallel to the y axis, at distance $x = 3.00$ m. In unit-vector notation, what is the velocity of the electron when it hits the screen?

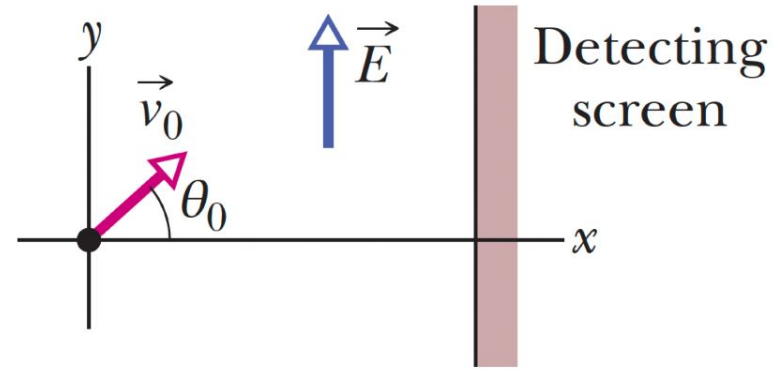


Figure 22-61 Problem 54.

53 • An electron has an initial velocity of 2.00×10^6 m/s in the $+x$ direction. It enters a region that has a uniform electric field $\vec{E} = (300 \text{ N/C})\hat{j}$. (a) Find the acceleration of the electron. (b) How long does it take for the electron to travel 10.0 cm in the $+x$ direction in the region that has the field? (c) Through what angle, and in what direction, is the electron deflected while traveling the 10.0 cm in the x direction?

54 •• An electron is released from rest in a weak electric field given by $\vec{E} = -1.50 \times 10^{-10} \text{ N/C}\hat{j}$. After the electron has traveled a vertical distance of $1.0 \mu\text{m}$, what is its speed? (Do not neglect the gravitational force on the electron.)

55 •• A 2.00-g charged particle is released from rest in a region that has a uniform electric field $\vec{E} = (300 \text{ N/C})\hat{i}$. After traveling a distance of 0.500 m in this region, the particle has a kinetic energy of 0.120 J. Determine the charge of the particle.

56 •• A charged particle leaves the origin with a speed of 3.00×10^6 m/s at an angle of 35° above the x axis. A uniform electric field, given by $\vec{E} = -E_0\hat{j}$, exists throughout the region. Find E_0 such that the particle will cross the x axis at $x = 1.50$ cm if the particle is (a) an electron and (b) a proton.

57 •• An electron starts at the position shown in Figure 21-39 with an initial speed $v_0 = 5.00 \times 10^6$ m/s at 45° to the x axis. The electric field is in the $+y$ direction and has a magnitude of 3.50×10^3 N/C. The black lines in the figure are charged metal plates. On which plate and at what location will the electron strike? **SSM**

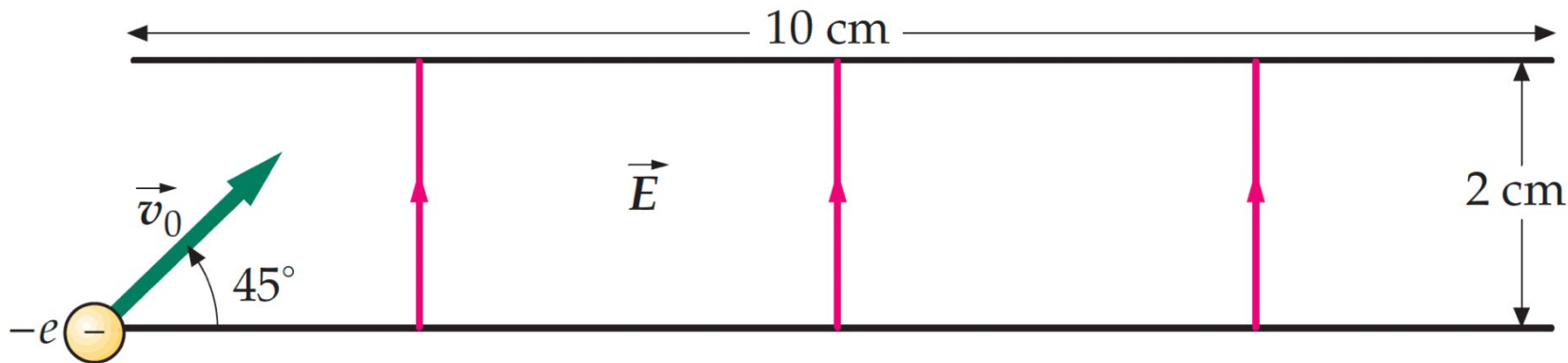
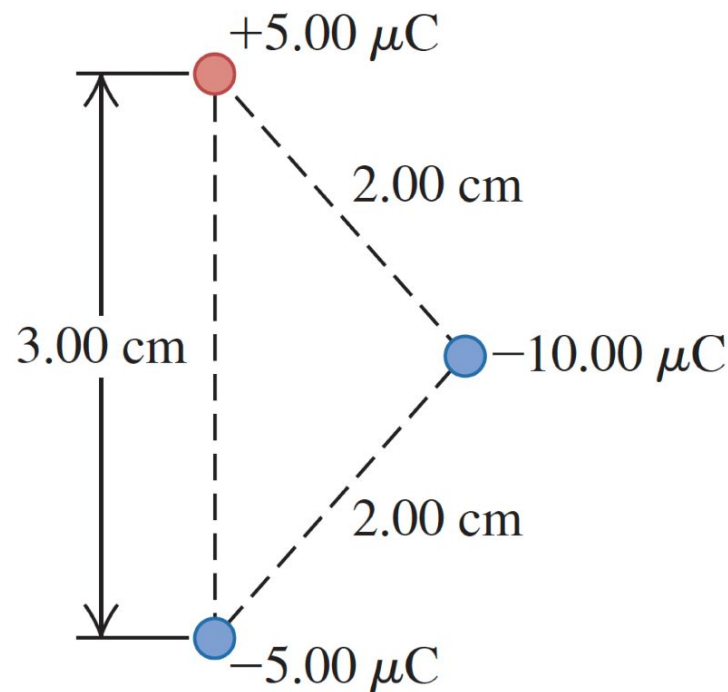


FIGURE 21-39 Problem 57

Electric Dipole in an Electric Field

21.55 • Three charges are at the corners of an isosceles triangle as shown in **Fig. E21.55**. The $\pm 5.00 \mu\text{C}$ charges form a dipole. (a) Find the force (magnitude and direction) the $-10.00 \mu\text{C}$ charge exerts on the dipole. (b) For an axis perpendicular to the line connecting the $\pm 5.00 \mu\text{C}$ charges at the midpoint of this line, find the torque (magnitude and direction) exerted on the dipole by the $-10.00 \mu\text{C}$ charge.

Figure **E21.55**



33. || A point charge Q is distance r from a dipole consisting of charges $\pm q$ separated by distance s . The dipole is initially oriented so that Q is in the plane bisecting the dipole. Immediately after the dipole is released, what are (a) the magnitude of the force and (b) the magnitude of the torque on the dipole? You can assume $r \gg s$.

34. | The permanent electric dipole moment of a particular molecule is $1.1 \times 10^{-30} \text{ C m}$. What is the maximum possible torque on the molecule in a $8.0 \times 10^8 \text{ N/C}$ field?

35. || An ammonia molecule (NH_3) has a permanent electric dipole moment $5.0 \times 10^{-30} \text{ C m}$. A proton is 2.0 nm from the molecule in the plane that bisects the dipole. What is the magnitude of the torque on the molecule?

21.56 • An electric dipole with a dipole moment of magnitude p is placed at various orientations in an electric field $\vec{E}$ that is directed to the left. (a) What orientation of the dipole will result in maximum torque directed into the page? What then is the electric potential energy? (b) What orientation of the dipole will give zero torque and maximum electric potential energy? What type of equilibrium is this: stable, unstable, or neutral?

Electric Flux

Module 23-1 Electric Flux

•1 **SSM** The square surface shown in Fig. 23-30 measures 3.2 mm on each side. It is immersed in a uniform electric field with magnitude $E = 1800 \text{ N/C}$ and with field lines at an angle of $\theta = 35^\circ$ with a normal to the surface, as shown. Take that normal to be directed “outward,” as though the surface were one face of a box. Calculate the electric flux through the surface.

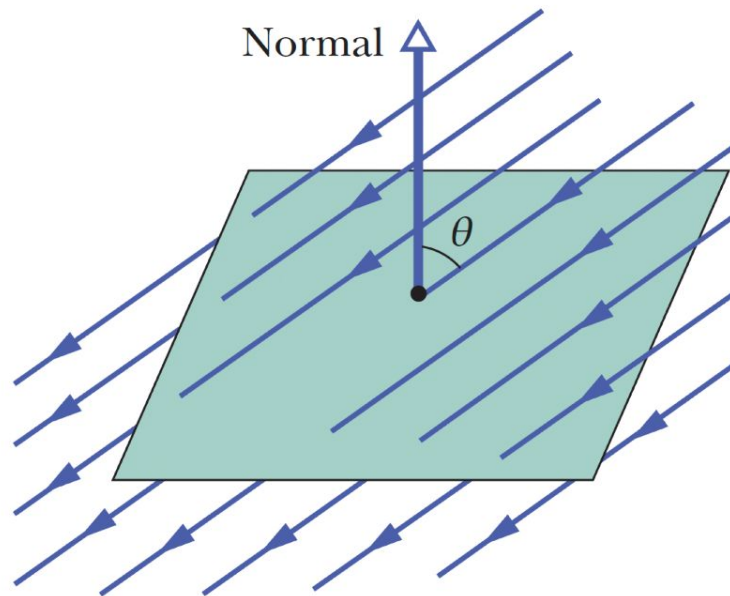
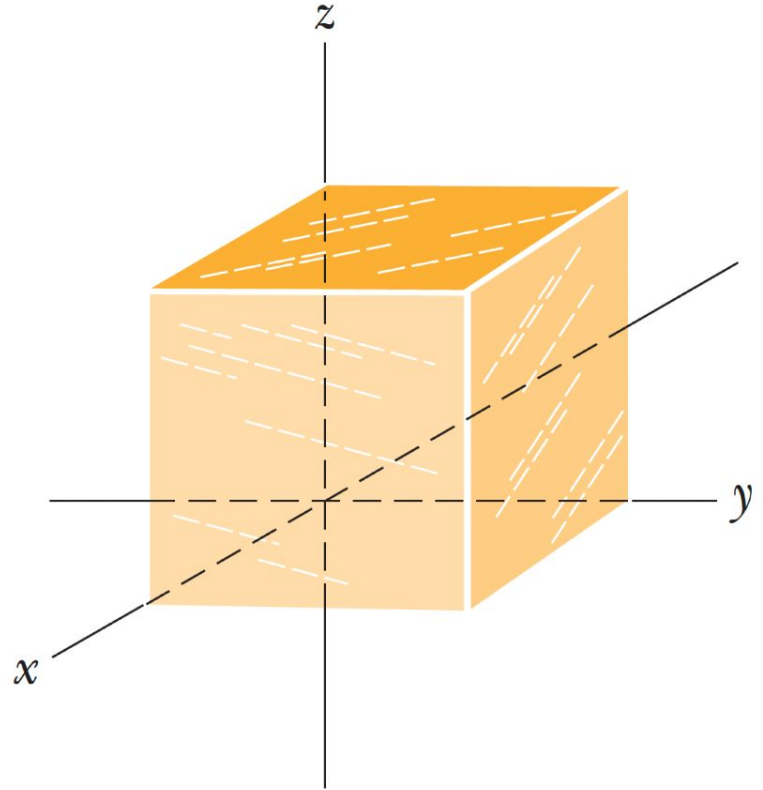


Figure 23-30 Problem 1.

••2 An electric field given by $\vec{E} = 4.0\hat{i} - 3.0(y^2 + 2.0)\hat{j}$ pierces a Gaussian cube of edge length 2.0 m and positioned as shown in Fig. 23-7. (The magnitude E is in newtons per coulomb and the position x is in meters.) What is the electric flux through the (a) top face, (b) bottom face, (c) left face, and (d) back face? (e) What is the net electric flux through the cube?



Module 23-2 Gauss' Law

•4 In Fig. 23-32, a butterfly net is in a uniform electric field of magnitude $E = 3.0 \text{ mN/C}$. The rim, a circle of radius $a = 11 \text{ cm}$, is aligned perpendicular to the field. The net contains no net charge. Find the electric flux through the netting.

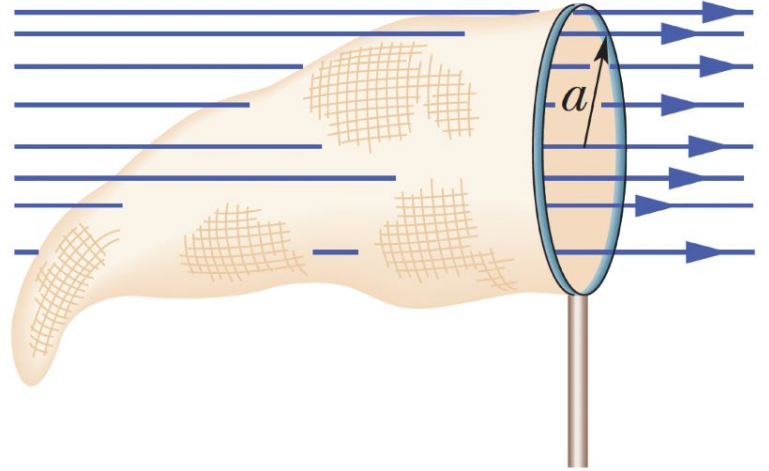


Figure 23-32 Problem 4.

●●11  Figure 23-35 shows a closed Gaussian surface in the shape of a cube of edge length 2.00 m, with one corner at $x_1 = 5.00$ m, $y_1 = 4.00$ m. The cube lies in a region where the electric field vector is given by $\vec{E} = -3.00\hat{i} - 4.00y^2\hat{j} + 3.00\hat{k}$ N/C, with y in meters. What is the net charge contained by the cube?

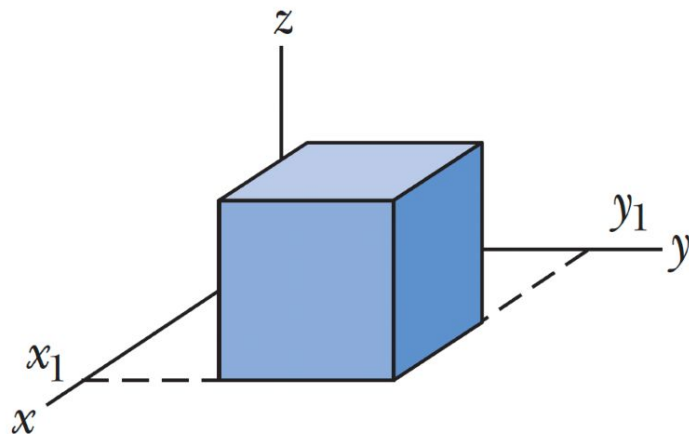


Figure 23-35 Problem 11.

••9 **ILW** Fig. 23-31 shows a Gaussian surface in the shape of a cube with edge length 1.40 m. What are (a) the net flux Φ through the surface and (b) the net charge q_{enc} enclosed by the surface if $\vec{E} = (3.00y\hat{j})$ N/C, with y in meters? What are (c) Φ and (d) q_{enc} if $\vec{E} = [-4.00\hat{i} + (6.00 + 3.00y)\hat{j}]$ N/C?

••10 Figure 23-34 shows a closed Gaussian surface in the shape of a cube of edge length 2.00 m. It lies in a region where the nonuniform electric field is given by $\vec{E} = (3.00x + 4.00)\hat{i} + 6.00\hat{j} + 7.00\hat{k}$ N/C, with x in meters. What is the net charge contained by the cube?

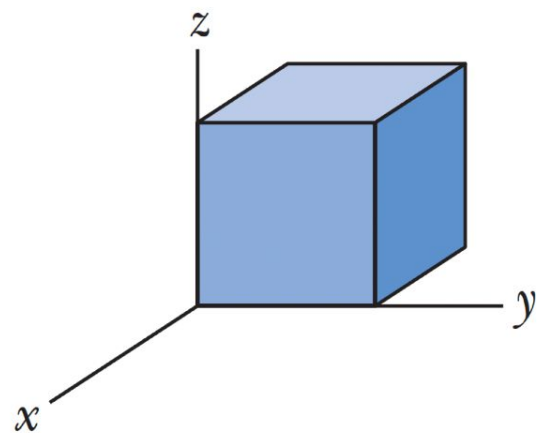


Figure 23-34
Problem 10.

7 •• An electric dipole is completely inside a closed imaginary surface and there are no other charges. True or false:

- (a) The electric field is zero everywhere on the surface.
- (b) The electric field is normal to the surface everywhere on the surface.
- (c) The electric flux through the surface is zero.
- (d) The electric flux through the surface could be positive or negative.
- (e) The electric flux through a portion of the surface might not be zero. **SSM**

Gauss's Law: Spherical Symmetry

●●49 In Fig. 23-54, a solid sphere of radius $a = 2.00$ cm is concentric with a spherical conducting shell of inner radius $b = 2.00a$ and outer radius $c = 2.40a$. The sphere has a net uniform charge $q_1 = +5.00$ fC; the shell has a net charge $q_2 = -q_1$. What is the magnitude of the electric field at radial distances (a) $r = 0$, (b) $r = a/2.00$, (c) $r = a$, (d) $r = 1.50a$, (e) $r = 2.30a$, and (f) $r = 3.50a$? What is the net charge on the (g) inner and (h) outer surface of the shell?

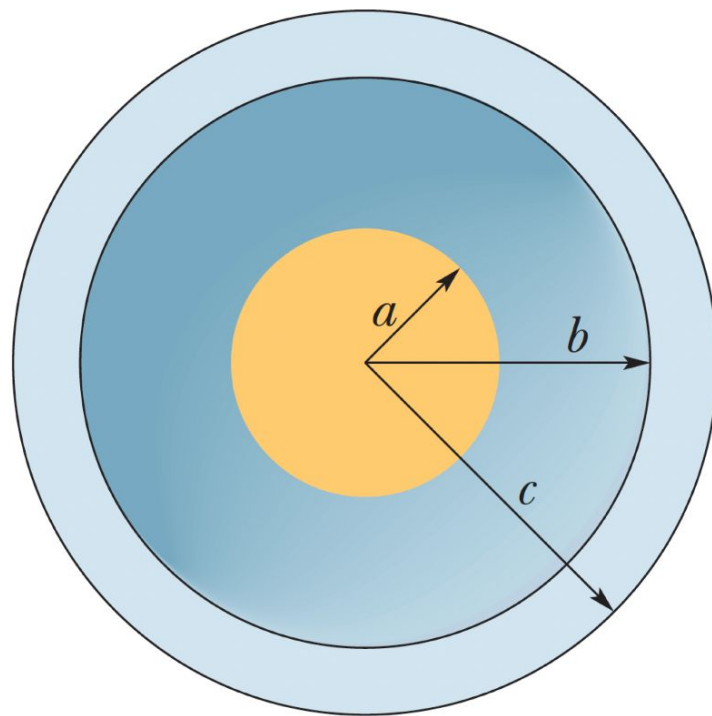


Figure 23-54 Problem 49.

•44 Figure 23-52 gives the magnitude of the electric field inside and outside a sphere with a positive charge distributed uniformly throughout its volume. The scale of the vertical axis is set by $E_s = 5.0 \times 10^7 \text{ N/C}$. What is the charge on the sphere?

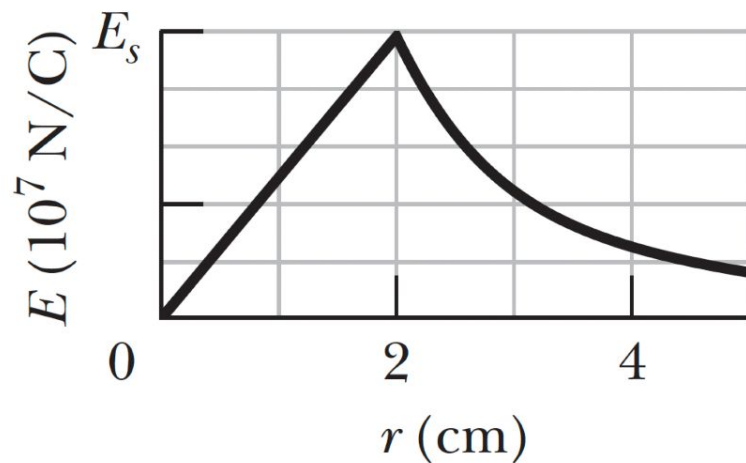


Figure 23-52 Problem 44.

•45 Two charged concentric spherical shells have radii 10.0 cm and 15.0 cm. The charge on the inner shell is $4.00 \times 10^{-8} \text{ C}$, and that on the outer shell is $2.00 \times 10^{-8} \text{ C}$. Find the electric field (a) at $r = 12.0 \text{ cm}$ and (b) at $r = 20.0 \text{ cm}$.

57 A thin-walled metal spherical shell has radius 25.0 cm and charge 2.00×10^{-7} C. Find E for a point (a) inside the shell, (b) just outside it, and (c) 3.00 m from the center.

58 A uniform surface charge of density 8.0 nC/m^2 is distributed over the entire xy plane. What is the electric flux through a spherical Gaussian surface centered on the origin and having a radius of 5.0 cm?

38 • A nonconducting thin spherical shell of radius 6.00 cm has a uniform surface charge density of 9.00 nC/m^2 . (a) What is the total charge on the shell? Find the electric field at the following distances from the sphere's center: (b) 2.00 cm, (c) 5.90 cm, (d) 6.10 cm, and (e) 10.0 cm.

39 • • A nonconducting sphere of radius 6.00 cm has a uniform volume charge density of 450 nC/m^3 . (a) What is the total charge on the sphere? Find the electric field at the following distances from the sphere's center: (b) 2.00 cm, (c) 5.90 cm, (d) 6.10 cm, and (e) 10.0 cm. **SSM**

Gauss's Law: Cylindrical Symmetry

•**24** Figure 23-40 shows a section of a long, thin-walled metal tube of radius $R = 3.00$ cm, with a charge per unit length of $\lambda = 2.00 \times 10^{-8}$ C/m. What is the magnitude E of the electric field at radial distance (a) $r = R/2.00$ and (b) $r = 2.00R$? (c) Graph E versus r for the range $r = 0$ to $2.00R$.

•**25** **SSM** An infinite line of charge produces a field of magnitude 4.5×10^4 N/C at distance 2.0 m. Find the linear charge density.

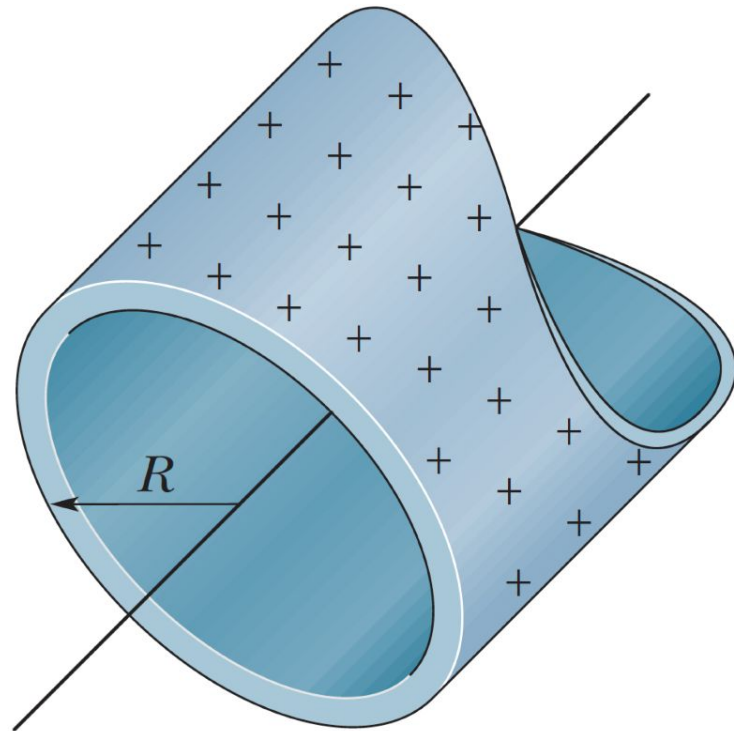


Figure 23-40 Problem 24.

••27  A long, straight wire has fixed negative charge with a linear charge density of magnitude 3.6 nC/m . The wire is to be enclosed by a coaxial, thin-walled nonconducting cylindrical shell of radius 1.5 cm . The shell is to have positive charge on its outside surface with a surface charge density σ that makes the net external electric field zero. Calculate σ .

••28  A charge of uniform linear density 2.0 nC/m is distributed along a long, thin, nonconducting rod. The rod is coaxial with a long conducting cylindrical shell (inner radius $= 5.0 \text{ cm}$, outer radius $= 10 \text{ cm}$). The net charge on the shell is zero. (a) What is the magnitude of the electric field 15 cm from the axis of the shell? What is the surface charge density on the (b) inner and (c) outer surface of the shell?

••30 In Fig. 23-43, short sections of two very long parallel lines of charge are shown, fixed in place, separated by $L = 8.0$ cm. The uniform linear charge densities are $+6.0 \mu\text{C}/\text{m}$ for line 1 and $-2.0 \mu\text{C}/\text{m}$ for line 2. Where along the x axis shown is the net electric field from the two lines zero?

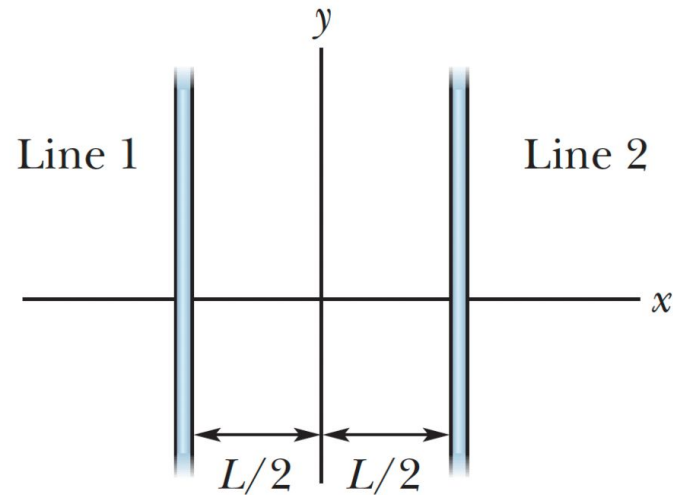


Figure 23-43 Problem 30.

••31 **ILW** Two long, charged, thin-walled, concentric cylindrical shells have radii of 3.0 and 6.0 cm. The charge per unit length is $5.0 \times 10^{-6} \text{ C}/\text{m}$ on the inner shell and $-7.0 \times 10^{-6} \text{ C}/\text{m}$ on the outer shell. What are the (a) magnitude E and (b) direction (radially inward or outward) of the electric field at radial distance $r = 4.0$ cm? What are (c) E and (d) the direction at $r = 8.0$ cm?

Gauss's Law: Planar Symmetry

69 Figure 23-59 shows, in cross section, three infinitely large nonconducting sheets on which charge is uniformly spread. The surface charge densities are $\sigma_1 = +2.00 \mu\text{C}/\text{m}^2$, $\sigma_2 = +4.00 \mu\text{C}/\text{m}^2$, and $\sigma_3 = -5.00 \mu\text{C}/\text{m}^2$, and distance $L = 1.50 \text{ cm}$. In unit-vector notation, what is the net electric field at point P ?

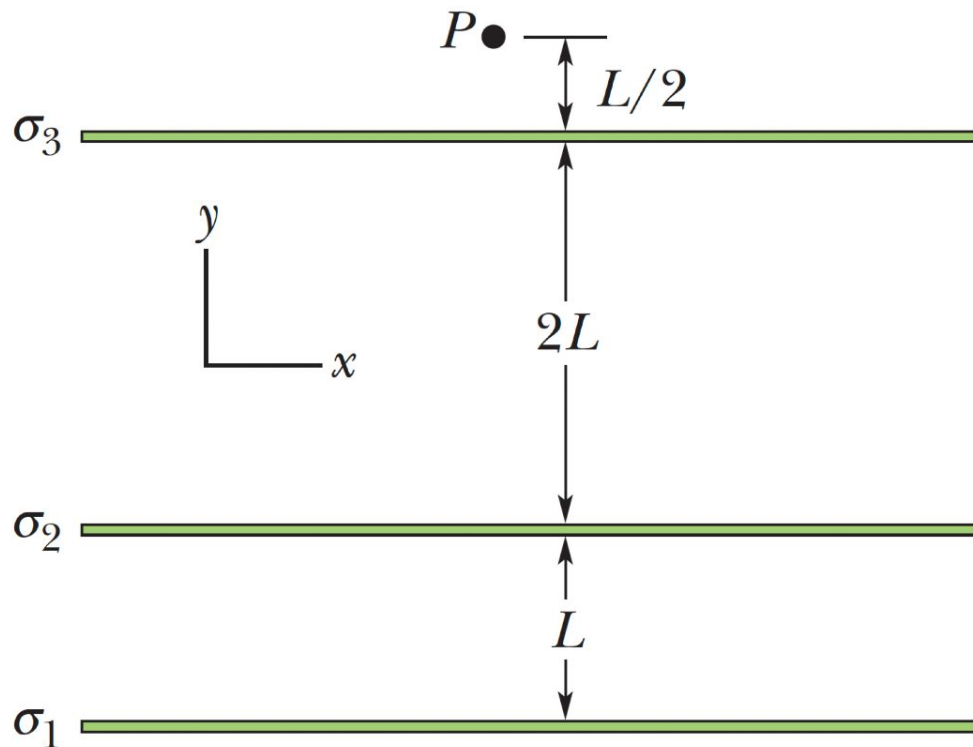


Figure 23-59 Problem 69.

Electric Potential Energy & Potential: Discrete Sources

7 Figure 24-30 shows a system of three charged particles. If you move the particle of charge $+q$ from point A to point D , are the following quantities positive, negative, or zero: (a) the change in the electric potential energy of the three-particle system, (b) the work done by the net electric force on the particle you moved (that is, the net force due to the other two particles), and (c) the work done by your force? (d) What are the answers to (a) through (c) if, instead, the particle is moved from B to C ?

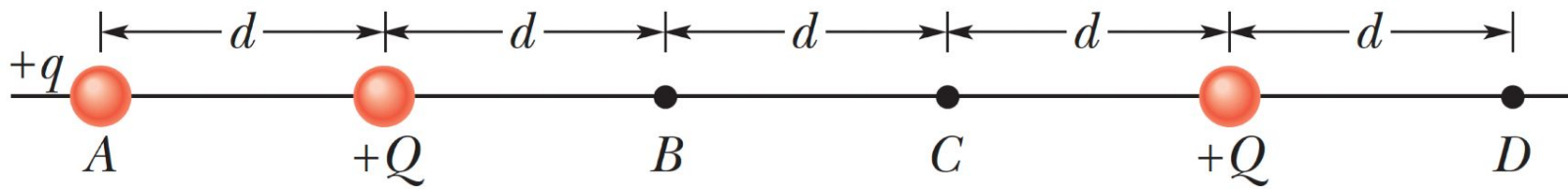


Figure 24-30 Questions 7 and 8.

8 In the situation of Question 7, is the work done by your force positive, negative, or zero if the particle is moved (a) from A to B , (b) from A to C , and (c) from B to D ? (d) Rank those moves according to the magnitude of the work done by your force, greatest first.

23.1 ●● A point charge $q_1 = +2.10 \mu\text{C}$ is held stationary at the origin. A second point charge $q_2 = -4.60 \mu\text{C}$ moves from the point $x = 0.150 \text{ m}$, $y = 0$ to the point $x = 0.250 \text{ m}$, $y = 0.270 \text{ m}$. How much work is done by the electric force on q_2 ?

23.2 ● A point charge q_1 is held stationary at the origin. A second charge q_2 is placed at point a , and the electric potential energy of the pair of charges is $+5.4 \times 10^{-8} \text{ J}$. When the second charge is moved to point b , the electric force on the charge does $-1.9 \times 10^{-8} \text{ J}$ of work. What is the electric potential energy of the pair of charges when the second charge is at point b ?

23.4 ●● (a) How much work would it take to push two protons very slowly from a separation of 2.00×10^{-10} m (a typical atomic distance) to 3.00×10^{-15} m (a typical nuclear distance)? (b) If the protons are both released from rest at the closer distance in part (a), how fast are they moving when they reach their original separation?

23.7 ●● Two protons, starting several meters apart, are aimed directly at each other with speeds of 2.00×10^5 m/s, measured relative to the earth. Find the maximum electric force that these protons will exert on each other.

23.8 ●● Three equal $1.40 \mu\text{C}$ point charges are placed at the corners of an equilateral triangle with sides 0.300 m long. What is the potential energy of the system? (Take as zero the potential energy of the three charges when they are infinitely far apart.)

VP23.2.1 An electron (charge $-e = -1.60 \times 10^{-19}$ C) is at rest at a distance 1.00×10^{-10} m from the center of a nucleus of lead (charge $+82e = +1.31 \times 10^{-17}$ C). You may treat both objects as point charges. (a) How much work would you have to do to move the electron to a distance 5.00×10^{-10} m from the center of the lead nucleus? (b) If you release the electron from the position in (a), what will be its kinetic energy when it is a distance 8.00×10^{-12} m from the center of the lead nucleus? (You can assume that the lead nucleus remains at rest, since it is much more massive than the electron.)

VP23.2.2 In a nuclear physics experiment, a proton (mass 1.67×10^{-27} kg, charge $+e = +1.60 \times 10^{-19}$ C) is fired directly at a target nucleus of unknown charge. (You can treat both objects as point charges, and assume that the nucleus remains at rest.) When it is far from its target, the proton has speed 2.50×10^6 m/s. The proton comes momentarily to rest at a distance 5.29×10^{-13} m from the center of the target nucleus, then flies back in the direction from which it came. (a) What is the electric potential energy of the proton and nucleus when they are 5.29×10^{-13} m apart? (b) What is the charge of the target nucleus?

VP23.2.3 Two point charges are at fixed positions on the y -axis: $q_1 = +e$ at $y = 0$ and $q_2 = -e$ at $y = a$. Find (a) the work you must do to bring a third charge $q_3 = -e$ from infinity to being at rest at $y = -2a$ and (b) the total potential energy of the system of three charges.

23.12 • An object with charge $q = -6.00 \times 10^{-9} \text{ C}$ is placed in a region of uniform electric field and is released from rest at point A . After the charge has moved to point B , 0.500 m to the right, it has kinetic energy $3.00 \times 10^{-7} \text{ J}$. (a) If the electric potential at point A is $+30.0 \text{ V}$, what is the electric potential at point B ? (b) What are the magnitude and direction of the electric field?

23.13 • A small particle has charge $-5.70 \mu\text{C}$ and mass $2.70 \times 10^{-4} \text{ kg}$. It moves from point A , where the electric potential is $V_A = +270 \text{ V}$, to point B , where the electric potential is $V_B = +830 \text{ V}$. The electric force is the only force acting on the particle. The particle has speed 5.90 m/s at point A . What is its speed at point B ? Is it moving faster or slower at B than at A ? Explain.

23.14 • A particle with charge $+4.20 \text{ nC}$ is in a uniform electric field $\vec{E}$ directed to the left. The charge is released from rest and moves to the left; after it has moved 6.00 cm , its kinetic energy is $+2.20 \times 10^{-6} \text{ J}$. What are (a) the work done by the electric force, (b) the potential of the starting point with respect to the end point, and (c) the magnitude of $\vec{E}$?

23.24 ●● A small sphere with charge $q = -5.00 \mu\text{C}$ is moving in a uniform electric field that has no y - or z -component. The only force on the sphere is the force exerted by the electric field. Point A is on the x -axis at $x = -0.400 \text{ m}$, and point B is at the origin. At point A the sphere has kinetic energy $K_A = 8.00 \times 10^{-4} \text{ J}$, and at point B its kinetic energy is $K_B = 3.00 \times 10^{-4} \text{ J}$. (a) What is the potential difference $V_{AB} = V_A - V_B$? Which point, A or B , is at higher potential? (b) What are the magnitude and direction of the electric field?

23.25 ●● Identical point charges q_1 and q_2 each have positive charge $+6.00 \mu\text{C}$. Charge q_1 is held fixed on the x -axis at $x = +0.400 \text{ m}$, and q_2 is held fixed on the x -axis at $x = -0.400 \text{ m}$. A small sphere has charge $Q = -0.200 \mu\text{C}$ and mass 12.0 g . The sphere is initially very far from the origin. It is released from rest and moves along the y -axis toward the origin. (a) As the sphere moves from very large y to $y = 0$, how much work is done on it by the resultant force exerted by q_1 and q_2 ? (b) If the only force acting on the sphere is the force exerted by the point charges, what is its speed when it reaches the origin?

6 Figure 24-29 shows four arrangements of charged particles, all the same distance from the origin. Rank the situations according to the net electric potential at the origin, most positive first. Take the potential to be zero at infinity.

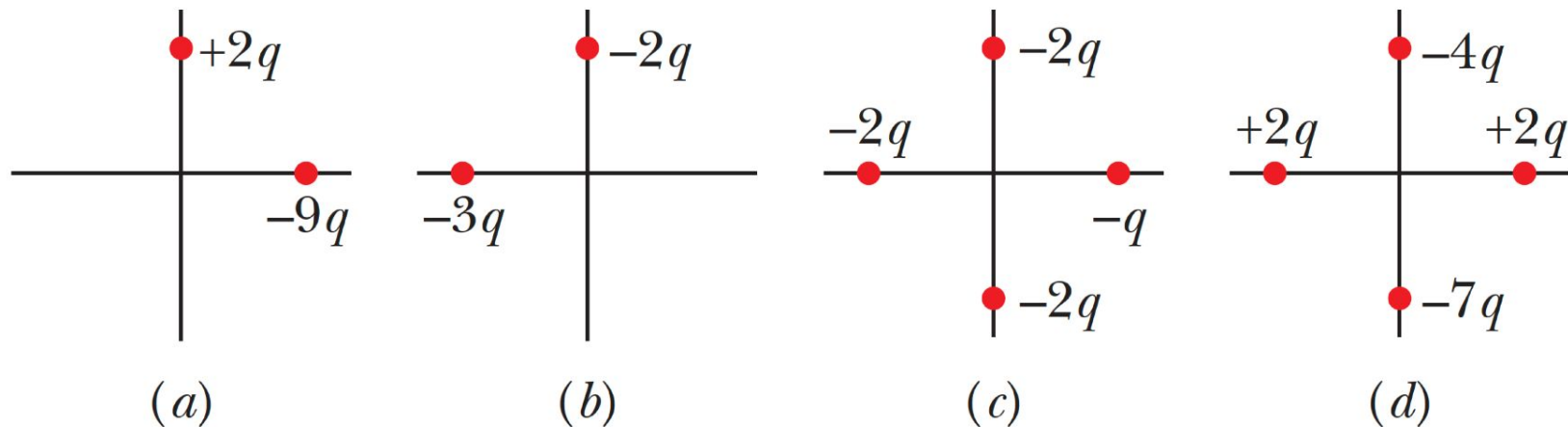


Figure 24-29 Question 6.

••17  In Fig. 24-38, what is the net electric potential at point P due to the four particles if $V = 0$ at infinity, $q = 5.00$ fC, and $d = 4.00$ cm?

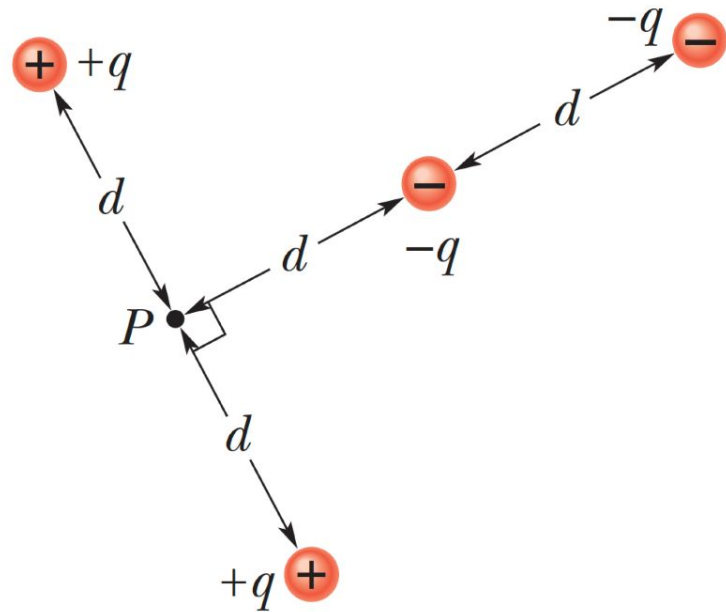


Figure 24-38 Problem 17.

••19 In Fig. 24-40, particles with the charges $q_1 = +5e$ and $q_2 = -15e$ are fixed in place with a separation of $d = 24.0$ cm. With electric potential defined to be $V = 0$ at infinity, what are the finite (a) positive and (b) negative values of x at which the net electric potential on the x axis is zero?

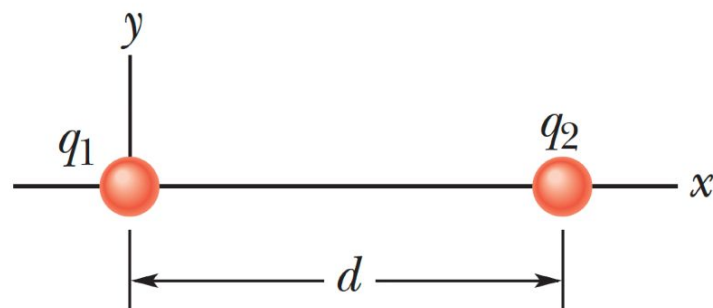


Figure 24-40 Problems 19 and 20.

••20 Two particles, of charges q_1 and q_2 , are separated by distance d in Fig. 24-40. The net electric field due to the particles is zero at $x = d/4$. With $V = 0$ at infinity, locate (in terms of d) any point on the x axis (other than at infinity) at which the electric potential due to the two particles is zero.

•14 Consider a particle with charge $q = 1.0 \mu\text{C}$, point A at distance $d_1 = 2.0 \text{ m}$ from q , and point B at distance $d_2 = 1.0 \text{ m}$. (a) If A and B are diametrically opposite each other, as in Fig. 24-36a, what is the electric potential difference $V_A - V_B$? (b) What is that electric potential difference if A and B are located as in Fig. 24-36b?

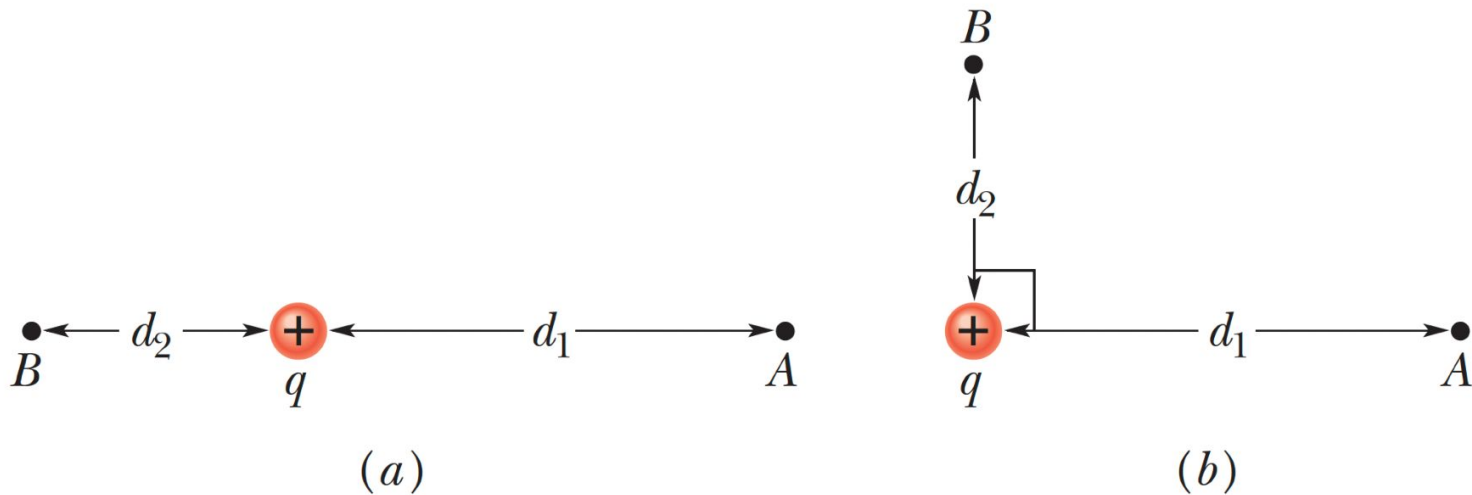


Figure 24-36 Problem 14.

Electric Potential Energy and Potential: Continuous Source

•4 Two large, parallel, conducting plates are 12 cm apart and have charges of equal magnitude and opposite sign on their facing surfaces. An electric force of 3.9×10^{-15} N acts on an electron placed anywhere between the two plates. (Neglect fringing.) (a) Find the electric field at the position of the electron. (b) What is the potential difference between the plates?

•5 **SSM** An infinite nonconducting sheet has a surface charge density $\sigma = 0.10 \mu\text{C}/\text{m}^2$ on one side. How far apart are equipotential surfaces whose potentials differ by 50 V?

•6 When an electron moves from A to B along an electric field line in Fig. 24-34, the electric field does $3.94 \times 10^{-19} \text{ J}$ of work on it. What are the electric potential differences (a) $V_B - V_A$, (b) $V_C - V_A$, and (c) $V_C - V_B$?

••7 The electric field in a region of space has the components $E_y = E_z = 0$ and $E_x = (4.00 \text{ N/C})x$. Point A is on the y axis at $y = 3.00 \text{ m}$, and point B is on the x axis at $x = 4.00 \text{ m}$. What is the potential difference $V_B - V_A$?

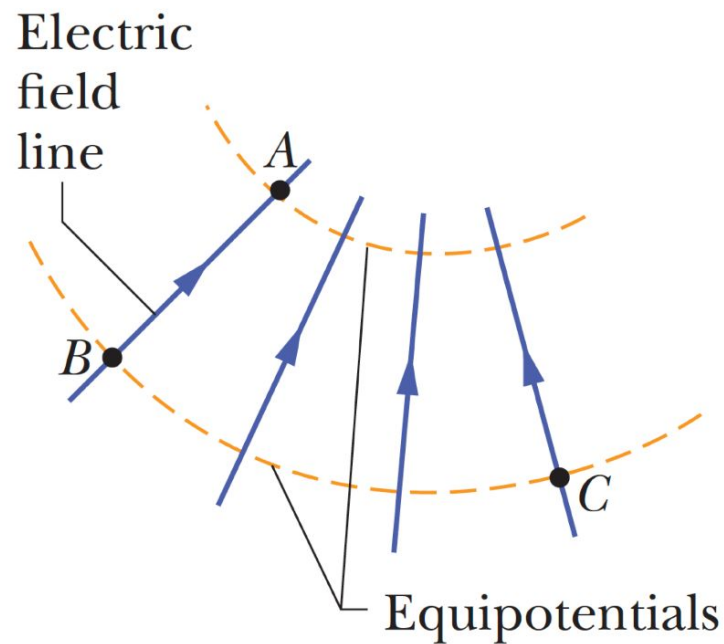


Figure 24-34 Problem 6.

What is the potential difference $V_B - V_A$?

•**34** Two large parallel metal plates are 1.5 cm apart and have charges of equal magnitudes but opposite signs on their facing surfaces. Take the potential of the negative plate to be zero. If the potential halfway between the plates is then +5.0 V, what is the electric field in the region between the plates?

•**35** The electric potential at points in an xy plane is given by $V = (2.0 \text{ V/m}^2)x^2 - (3.0 \text{ V/m}^2)y^2$. In unit-vector notation, what is the electric field at the point (3.0 m, 2.0 m)?

•**36** The electric potential V in the space between two flat parallel plates 1 and 2 is given (in volts) by $V = 1500x^2$, where x (in meters) is the perpendicular distance from plate 1. At $x = 1.3 \text{ cm}$, (a) what is the magnitude of the electric field and (b) is the field directed toward or away from plate 1?

●●57 **SSM** Identical $50\ \mu\text{C}$ charges are fixed on an x axis at $x = \pm 3.0\ \text{m}$. A particle of charge $q = -15\ \mu\text{C}$ is then released from rest at a point on the positive part of the y axis. Due to the symmetry of the situation, the particle moves along the y axis and has kinetic energy $1.2\ \text{J}$ as it passes through the point $x = 0, y = 4.0\ \text{m}$. (a) What is the kinetic energy of the particle as it passes through the origin? (b) At what negative value of y will the particle momentarily stop?

••59 In Fig. 24-60, a charged particle (either an electron or a proton) is moving rightward between two parallel charged plates separated by distance $d = 2.00$ mm. The plate potentials are $V_1 = -70.0$ V and $V_2 = -50.0$ V. The particle is slowing from an initial speed of 90.0 km/s at the left plate. (a) Is the particle an electron or a proton? (b) What is its speed just as it reaches plate 2?

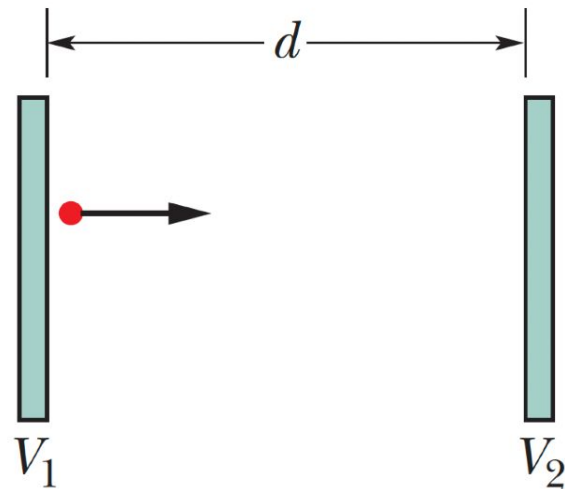


Figure 24-60
Problem 59.